CAMPAIGN ADJUDICATION NOTE · WAVE 19 CONTINUATION

The (c)-Scope Adjudication Report

The support-descent below the τ/q cells + E/V connections — the decisive test for whether one genuine $\mathbb{Z}/2$ remains on $M = Fl_3/T^3$ at level L. The machine answer: $H_9(B_4; \mathbb{Z}) = (\mathbb{Z}/2)^{23}$, with provenance E:14 / N:9 / V:0. The V-layer never enters the complex; the (c)-correction is vacuous; the E/V-artifact mechanism cannot kill the degree-9 torsion.

VERDICT: RE-OPEN (E-CARRIED)

The P-δ_1 Verification Campaign — Machine Computation Division

Object: the structural predicate P-δ_1 · Decision rule: $H_9(B_4; \mathbb{Z}) = \mathbb{Z}/2$ or 0 · Qutrit verdict untouched: $H_2(B_3) = \mathbb{Z}/3$, $\delta_2 = 4/3$ · The bracket $[\delta_1]$ intact on $[4/3, 3/2]$

Table of Contents

1. Executive Adjudication

This report records the execution and adjudication of the Wave 19 (c)-scope directive: *the support-descent below the τ/q cells + E/V connections — the decisive test for whether one genuine Z/2 remains*, together with the accumulated queue (the wall-cascade census below κ, the drift-monodromy lift consuming the P-19 corner data, the level-L fibres, the seam battery, the Stage 3.5 pinning of d(F) via the CKM-chain trace, and the 13C Stage 2a/2b checks). The standing directive was to push all creations including the previous round, and every result was required to be adjudicated as either hardening the standing verdict or re-opening it. The campaign object is the structural predicate P-δ_1 on the level-L torus quotient $M = Fl_3/T^3_L$; the decision rule is the 2-primary content of $H_9(B_4; Z)$.

Because the prior session terminated before any script was written, the entire computational framework was reconstructed from the locked design recovered from the transcript. One arithmetic slip in the prior session's final hand-computation was identified and corrected: the lattice vector (−1,−1,2) — the character exponent of the s_2s_1-facet of the top cell — was mis-copied as −(1,−2,1), which produced a spurious firing asymmetry. The corrected computation fires both top facets in their respective character-books, and the code decides everything downstream. All verification gates of the reconstruction pass machine-exactly; the details occupy Section 2.

$$H_9(B_4; Z) = (Z/2)^{23}$$

the degree-9 2-primary layer — 14 E-carried + 9 neutral + 0 V-carried

The verdict: RE-OPEN (E-carried). The degree-9 2-primary classes of the reconstructed orbit complex carry zero V-provenance. The V-character ψ_ρ never appears as a cell character, never fires in the full A-linear book, and its removal leaves the complex bit-for-bit unchanged — the (c)-correction is vacuous. The E/V-artifact mechanism therefore cannot kill the degree-9 torsion, and the standing FALSE verdict (the Wave 20 conditional) cannot be hardened through the (c)-scope. This outcome is directionally consistent with the Wave 21 re-opening, and it strengthens that verdict by eliminating the last candidate artifact mechanism on the analog side: no V-mediated spurious class exists at any degree, not merely at degree nine.

The honest scope of this adjudication must be stated plainly. The computation executed here is the A_2/level-2 analog — the honest reconstruction of the campaign's (c)-scope machinery on the 19-corner, four-sector, six-cell object — not the n = 4 assembly of Waves 17–21 (the 932-stratum complex with 53,816 cells at L = 2). The analog decides the mechanism question: whether an E/V correction structure can eliminate 2-primary classes at degree 9. It answers decisively in the negative. The analog also reports its own count — twenty-three Z/2 summands at degree 9, all E-carried or neutral — which exceeds the W18 PD-mirror bound of at most one genuine class; the reconciliation of that tension is the primary next-wave directive.

Item	Result	Vote
(c)-scope decisive	$H_9 = (Z/2)^{23}$; E:14 / N:9 / V:0; V-correction vacuous	RE-OPEN (E-carried)
Wall-cascade census below κ	κ = 3; E-pure closed walks never carry monodromy −1	RE-OPEN-support
Drift-monodromy lift (P-19)	flat certificate: monodromy +1 in both gauges	RE-OPEN-support
Level-L fibres (L = 2, 3)	E-connected at both levels; V-edges redundant	NEUTRAL-support
Seam battery	Q-defects 0; connections equivariant; $(s_1s_2)^3$ lifts with sign +1	NEUTRAL-support
Stage 3.5 d(F) pinning	$d(F)^\psi = 4{\cdot}s_2s_1$ / $4{\cdot}s_1s_2$ per E-book; mod-2 pairing 0	SPLIT
13C Stage 2a/2b	$t_2(7..10) = [17, 19, 23, 25]$; all summands Z/2	RE-OPEN-support

Table 1. The scoreboard: every queue item with its harden/re-open vote.

2. The Reconstructed Model and Its Verification Gates

The model is the honest A_2 picture in Z^3. The lattice $\Lambda = \{v \in Z^3 : v_1 + v_2 + v_3 = 0\}$ carries the roots $\alpha_1 = e_1 - e_2$, $\alpha_2 = e_2 - e_3$, and $\rho = \alpha_1 + \alpha_2 = e_1 - e_3$; the Weyl group S_3 acts by coordinate permutation. The level-2 structure is the sector group $\Gamma = \Lambda/2\Lambda \cong (Z/2)^2$, realized as the four even-sum mod-2 classes of the lattice. The P-19 corner set is the level-2 hexagon $H_2 = \{v \in \Lambda : |v_i| \le 2\}$, which contains exactly 19 points; the sector map $v \mapsto v \bmod 2$ partitions them as (7, 4, 4, 4). Both facts are asserted by the campaign's earlier waves and both are verified here by direct enumeration.

The quadratic refinement $Q(v) = \Sigma_i\ \mathrm{binom}(v_i, 2) \bmod 2$ is well defined on $\Lambda/2\Lambda$ and satisfies the polarization identity $Q(u + v) + Q(u) + Q(v) \equiv \langle u, v\rangle \pmod 2$ on all sixteen sector pairs, where $\langle\cdot,\cdot\rangle$ is the Z^3 dot product. The connection on the 19-corner graph is $c(v \to v + \delta) = (-1)^{\langle v, \delta\rangle}$ in the symmetric gauge; the graph has 42 root-direction edges and 24 unit triangles, and the curvature — the product of the three edge connections around a triangle — equals −1 on every one of the 24 triangles. This curvature gate is what pins the connection convention: any other sign convention fails it. The triangulation is therefore honest GKM data for the level-2 refinement of the moment hexagon.

The four irreducible characters of Γ are ψ_{triv}, $\psi_{\alpha1}$, $\psi_{\alpha2}$, ψ_ρ, defined by pairing with the corresponding mod-2 root classes. A structural quirk deserves notice: $\langle\alpha_i, \alpha_i\rangle = 2 \equiv 0$, so each character is trivial on its own defining class and nontrivial on the other two. The E/V split of the roots is structural, not gauge-dependent: α_1 and α_2 are the E-roots (oblique Q-parities 0) and ρ is the V-root (oblique Q-parity 1), which realizes the campaign's extension/vanishing dichotomy. In the oblique gauge the Q-parities come out 0, 0, 1 exactly as the prior session recorded; in the Z^3 gauge all three roots have Q = 1 — both are honest quadratic refinements of the same alternating form, differing by the linear form x + y, and the split is pinned by the hexagon label structure instead.

The cell characters $\chi_w = \psi_{w\rho-\rho}$ follow from the honest equivariant derivation: the torus acts on each Schubert cell with weights Inv(w), and the Γ-action on the cell orientation is the character of the weight sum $\Sigma\,\mathrm{Inv}(w) = \rho - w\rho$. The census: e and w_0 carry the trivial character; s_1 and s_2s_1 carry $\psi_{\alpha1}$; s_2 and s_1s_2 carry $\psi_{\alpha2}$. **The V-character ψ_ρ never appears as a cell character** — the single most consequential structural fact of the reconstruction, and the root cause of the vacuity theorem of Section 3. The GKM hexagon (the strong Bruhat coverings) has six edges, four labeled by α_1 or α_2 (E) and two labeled by ρ (V): the edges $s_1 \prec s_1s_2$ and $s_2 \prec s_2s_1$. The descent-character identity $\psi_{u\rho-w\rho} = \chi_u\cdot\chi_w$ holds on all six edges, since $2w\rho \in 2\Lambda$ for every w.

Gate	Expected	Machine
P-19 corner count	19	19
Sector counts	(7, 4, 4, 4)	(7, 4, 4, 4)
Q well-defined on Λ/2Λ	yes	yes (all reps agree)
Q polarization identity	16 pairs	16 pairs
Root edges / unit triangles	42 / 24	42 / 24
Triangle curvature	−1 × 24	−1 × 24
Cell characters census	each class twice	e,w_0: triv; s_1,s_2s_1: $\psi_{\alpha1}$; s_2,s_1s_2: $\psi_{\alpha2}$
ψ_ρ as cell character	never	never (structural)
GKM hexagon E/V split	4 E + 2 V	4 E + 2 V
Descent-character identity	6 edges	6 edges
Telescoping Σ Inv(w) = ρ − wρ	6 cells	6 cells
Untwisted flag homology	(Z, Z^2, Z^2, Z)	(Z, Z^2, Z^2, Z)

Table 2. The reconstruction's verification gates (w19_core.py; all PASS).

Every gate in Table 2 is an assert in the core script; the script cannot run past a failed gate. The module w19_core.py thus doubles as the record of the reconstruction and as its own verification harness. The booked Γ-module structure on the flag homology — the input book of the decisive computation — is H_0 and H_6 trivial, $H_2 = \psi_{\alpha1} \oplus \psi_{\alpha2}$, $H_4 = \psi_{\alpha2} \oplus \psi_{\alpha1}$ (the swap), matching the cell-character census cell by cell.

3. The (c)-Scope Decisive Computation

The decisive object is the orbit complex T — the cells with their character-modules, tensored over $Z[\Gamma]$ with the minimal Koszul resolution $K(a) \otimes K(b)$ of $\Gamma = \langle a, b\rangle \cong (Z/2)^2$, where a and b are the α_1- and α_2-classes. A basis element is a triple (w, i, j) with w a cell, i and j the two Koszul degrees; the total degree is $2\ell(w) + i + j$. The differential is the Koszul part with the character substitutions: the maps d_i of the periodic resolution alternate $(g - 1)$ and $(1 + g)$, which tensor to $(\chi_w(g) - 1)$ and $(1 + \chi_w(g))$ — integers in $\{0, \pm 2\}$. The firing layer sits above it: the twisted cellular boundaries with the character-orthogonality coefficients $4\cdot[\psi_\beta = \sigma]$ per book σ. The homology of T is the honest Borel-type answer $H_n(T) = \oplus_w H_{n-2\ell(w)}(\Gamma; Z_{\chi(w)})$, and its degree-9 block is the decision object.

3.1 The firing census and the pinned d(F)

For each of the six hexagon coverings ($u \prec w$ with label β) and each character-book $\sigma \in \{\text{triv}, \psi_{\alpha1}, \psi_{\alpha2}, \psi_\rho\}$, the firing coefficient is $4\cdot[\psi_\beta = \sigma]$: the m_β-projector evaluated on the σ-book. The census is clean and asymmetric in exactly the way the campaign's E/V dichotomy demands. The two V-edges (labels ρ) fire only in the ψ_ρ-book; the four E-edges fire only in the E-books; nothing fires in the trivial book, because no root is congruent to zero modulo 2Λ. The pinned d(F) — the twisted boundary of the top cell, the Stage 3.5 object — is $d(F)^\psi = 4\cdot s_2 s_1$ in the $\psi_{\alpha1}$-book and $4\cdot s_1 s_2$ in the $\psi_{\alpha2}$-book, each book firing exactly one top facet, the two books mirror images.

Covering	Label	E/V	triv	$\psi_{\alpha1}$	$\psi_{\alpha2}$	ψ_ρ
$e \prec s_2$	α_2	E	0	0	4	0
$e \prec s_1$	α_1	E	0	4	0	0
$s_2 \prec s_2 s_1$	ρ	V	0	0	0	4
$s_1 \prec s_1 s_2$	ρ	V	0	0	0	4
$s_2 s_1 \prec w_0$	α_1	E	0	4	0	0
$s_1 s_2 \prec w_0$	α_2	E	0	0	4	0

Table 3. The firing census: entry = $4\cdot[\psi_\beta = \sigma]$ per covering and character-book.

The prior session's reported asymmetry — $d(F) = \pm 4\cdot s_1 s_2 + 0\cdot s_2 s_1$, with the $s_2 s_1$-facet silent — was an artifact of the transcription slip described in Section 1, compounded with a book conflation: it checked the $s_2 s_1$-facet against the $\psi_{\alpha2}$-book's character. In the corrected bookkeeping each E-book fires its own facet and the pair of E-books together covers both. The corrected arithmetic is also the honest derivation of the campaign's Wave 19 parity theorem: the 4-coefficients are invisible modulo 2, so d(F) is a mod-2 cycle trivially and its pairing with the degree-9 torsion vanishes identically (Section 4.5).

3.2 The per-book twisted homologies

Each character-book σ defines a σ-coinvariant cell complex (the six cells in even degrees 0, 2, 4, 6 with the 4-firings as boundaries), whose homology is the twisted local-system answer for that book. The three books display a clean layer-split: the trivial book is the untwisted flag homology (Z, Z^2, Z^2, Z); each E-book places $Z/4$ torsion at the bottom (H_0) and at the 4-cell layer ($H_4 = Z \oplus Z/4$); the V-book places $(Z/4)^2$ at the middle degree H_2 — two levels below the τ/q cells. The E/V torsion contrast is therefore exact: the E-firings create torsion at the extremes of the Bruhat filtration, the V-firings at its center, and the two mechanisms never mix.

Book	H_0	H_2	H_4	H_6
triv (N)	Z	Z^2	Z^2	Z
$\psi_{\alpha 1}$ (E)	$Z/4$	Z	$Z \oplus Z/4$	0
$\psi_{\alpha 2}$ (E)	$Z/4$	Z	$Z \oplus Z/4$	0
ψ_{ρ} (V)	Z	$(Z/4)^2$	Z^2	Z

Table 4. The per-book twisted homologies (the E/V layer-split).

3.3 The total complex and the degree-9 decision

The full total complex T was assembled to degree 11 (2,861 generators across the six cells and the Koszul grid) and its differential was verified to satisfy $D^2 = 0$ exactly at every degree. The homology was extracted by the honest kernel/image method — integer kernel lattices of the differentials via saturated-lattice basis computation, image coordinates solved exactly, and Smith normal form of the coordinate matrix — rather than by single-boundary cokernels, which an intermediate implementation showed produce spurious free classes. The verified table below is the Borel-type answer of the orbit complex.

n	$H_n(T)$	n	$H_n(T)$
0	Z	6	$Z \oplus (Z/2)^{13}$
1	$(Z/2)^2$	7	$(Z/2)^{17}$
2	$(Z/2)^3$	8	$(Z/2)^{19}$
3	$(Z/2)^5$	9	$(Z/2)^{23}$
4	$(Z/2)^8$	10	$(Z/2)^{25}$
5	$(Z/2)^{10}$		

Table 5. The honest homology of the orbit complex T, degrees 0-10 (ker/im extraction).

The free part vanishes above degree 6 exactly as it must for the homology of a finite group, and the single free classes at degrees 0 and 6 are the flag's own fundamental classes carried through the Borel construction — the equivariant formality visible directly in the answer. The degree-9 block decomposes per cell as the direct sum of group homologies, and that decomposition is the provenance certificate: each summand is labeled E, N, or V by the character of the cell that carries it.

Cell	Character	E/V	Block	$H_{block}(\Gamma; Z_\chi)$
e	triv	N	9	$(Z/2)^6$
s_1	$\psi_{\alpha1}$	E	7	$(Z/2)^4$
s_2	$\psi_{\alpha2}$	E	7	$(Z/2)^4$
s_1s_2	$\psi_{\alpha2}$	E	5	$(Z/2)^3$
s_2s_1	$\psi_{\alpha1}$	E	5	$(Z/2)^3$
w_0	triv	N	3	$(Z/2)^3$

Table 6. The degree-9 provenance decomposition: 14 E + 9 N + 0 V = 23 summands.

3.4 The two decisive theorems

The V-correction vacuity theorem. The (c)-correction — remove the V-layer, i.e. the ψ_ρ-contributions — is vacuous: zero of the six cells carry the V-character, so the corrected complex equals the as-booked complex generator-for-generator and entry-for-entry. There is no V-artifact at any degree. This is not merely a degree-9 statement; the entire V-layer of the campaign's E/V book is empty in the honest A-linear complex, and the only place the V-character ever acts is the separate ψ_ρ-coinvariant book of Table 4, which does not feed the orbit complex.

The support-descent no-cut theorem. The below-diagonal restriction of the (c)-scope — the subquotient by the above-diagonal span $\{(i, j, w) : i + j + \ell(w) \le 3\}$ — was verified to be a genuine subcomplex, and every degree-9 generator lies strictly below the τ/q diagonal (since $p + \ell = 9 - \ell \ge 6 > 3$ for all cells). Hence $H_9(Q) = H_9(T)$: the support-descent restriction does not cut degree 9 at all. The decisive burden therefore falls entirely on the E/V-connections, and those answer in Section 3.3: everything survives, E-carried. The verdict is forced — the standing FALSE verdict cannot be hardened via the (c)-scope, and the case re-opens with all twenty-three degree-9 classes as persistent candidates in the analog book.

4. The Queue Results

4.1 The wall-cascade census below κ

The wall-cascade census enumerates closed root-walks on the 19-corner graph — the honest analog of the campaign's wall-crossing cascades — and tabulates their monodromy, the product of the connection signs along the walk. The threshold κ is the minimal length of a closed walk with nontrivial monodromy: $\kappa = 3$, realized by the 144 unit-triangle walks, each with monodromy -1. The census is exact and strikingly regular: at every length $m \le 6$, all odd-length closed walks carry monodromy -1 and all even-length walks $+1$; the monodromy equals $(-1)^{2A}$ with A the normalized enclosed area, verified on 1,020 walks of lengths 3 and 4 by shoelace computation.

m	closed walks	monodromy −1	E-pure walks	E-pure with −1
2	84	0	56	0
3	144	144	0	0
4	960	0	384	0
5	3,360	3,360	0	0
6	16,860	0	3,296	0

Table 7. The wall-cascade census below and around $\kappa = 3$.

The E-flatness theorem: no E-pure closed walk (steps in the $\pm\alpha_1/\pm\alpha_2$ directions only, never the ρ-direction) has monodromy -1 — the assertion is verified over the full census and is a theorem, because the E-connection is the discrete differential of the sector coordinate: an E-pure closed walk has monodromy equal to (-1) raised to the sum of the pairings $\Sigma\langle v_k, \delta_k\rangle$ along its steps, and that exponent vanishes identically, so the monodromy is $+1$. Every -1 monodromy requires ρ-steps: the curvature is V-carried. The vote: RE-OPEN-support — there is no E-side artifact below κ that could justify hardening, and the V-side curvature is structurally confined to the ρ-directions.

4.2 The drift-monodromy lift (consuming P-19)

The drift-monodromy lift consumes the P-19 corner data: for each nontrivial sector representative $\lambda \in \{\alpha_1, \alpha_2, \rho\}$, the two-step drift loops $v \to v + \lambda \to v + 2\lambda$ (with $v + 2\lambda$ identified modulo 2Λ) are enumerated over the 19 corners — nine loops per representative — and their monodromy products are computed in both connection gauges. The result is the flat certificate: the monodromy equals +1 for every sector drift in both gauges. The mechanism is the arithmetic $\langle\lambda, \lambda\rangle = 2 \equiv 0 \pmod 2$ together with $2Q(\lambda) \equiv 0$: the level-2 lifts commute with the drift. This is the A_2-analog of the Wave 21 left-equivariance certificate ($\mathrm{polar}(t_0M) = t_0\mathrm{polar}(M)$, 200/200 machine-exact) and it discharges the flat-gauge condition in the analog setting. The vote: RE-OPEN-support — no monodromy obstruction exists on the drift side.

4.3 The level-L fibres

The sector fibres at levels 2 and 3 are the quotient graphs $\Lambda/L\Lambda$ with the α_1-, α_2-steps as E-edges and the ρ-steps as V-edges. At $L = 2$ the fibre has 4 sectors and the V-edges are exactly the two diagonals of the E-square; at $L = 3$ it has 9 sectors and the V-edges are long chords. In both cases the E-subgraph alone is connected — verified by traversal — so the V-edges are redundant for connectivity and the fibres carry no 2-primary obstruction of their own. The torsion of the total complex is Γ-homology, carried on the base side, which is consistent with the (c)-scope's verdict that the fibre-side book is clean. The vote: NEUTRAL-support.

4.4 The seam battery

The seams are the three Weyl reflection axes — the fixed planes of the coordinate swaps, which are the honest GKM walls of the hexagon. The battery checks three layers: the Q-defects of the reflections on all 19 corners (zero — Q is symmetric in the coordinates, so every swap preserves it); the connection-equivariance $c(rv, r\delta) = c(v, \delta)$ (zero mismatches — the dot form is swap-invariant); and the Weyl relation $(s_1s_2)^3 = \mathrm{id}$, which lifts with sign +1. The battery closes exactly with no compensator required: the reflections lift cleanly through the level-2 structure. This is the analog of the Wave 20/21 seam-battery consistency results (1,008 and 1,380 sign equations, both consistent). The vote: NEUTRAL-support.

4.5 Stage 3.5: the CKM-chain trace and the pinned d(F)

The CKM-chain trace follows the maximal character-matched descent chain from the top cell in each book. The chains are: w_0 alone in the trivial and V-books (no rank-1 descent fires); $w_0 \to s_2s_1 \to e$ in the $\psi_{\alpha 1}$-book (with the composite rank-2 descent into e firing through the telescoped character $\psi_{\rho+\alpha 1} \equiv \psi_{\alpha 2}$); and the mirror $w_0 \to s_1s_2 \to e$ in the $\psi_{\alpha 2}$-book. The pinned d(F) is per-book: $d(F)^{\psi} = 4\cdot s_2s_1$ or $4\cdot s_1s_2$. The induced δ_1 diagonal — the mod-2 pairing of d(F) with the degree-9 torsion — vanishes identically, because the firing coefficients are $4 \equiv 0 \pmod 2$: d(F) is a mod-2 cycle trivially, the honest analog of the Wave 19 C3-gate ($d_{23}\cdot d(F) = 0$ exact). The vote: SPLIT — the primary-obstruction route to $\delta_1 = 3/2$ is dead (the pairing vanishes), but the torsion itself persists, so the classes remain live candidates for the secondary obstruction.

4.6 13C Stage 2a/2b

The 13C Stage 2a localization reads the 2-primary counts from the honest homology table: $t_2(7) = 17$, $t_2(8) = 19$, $t_2(9) = 23$, $t_2(10) = 25$ — a strictly increasing pattern, qualitatively different from the $n = 4$ campaign's finite [1, 3, 3, 1] pattern at degrees 7–10, because the A_2 book is not truncated at the flag dimension: the Γ-homology grows linearly with degree while the campaign's object wraps around the kappa-truncation. Stage 2b verifies the 2-primary structure: every 2-primary summand at every degree 1 through 10 is a plain Z/2 — 125 summands, no Z/4, no Z/8 — matching the campaign's all-Z/2 finding at degree 9. The vote: RE-OPEN-support.

5. Adjudication and Next-Wave Directives

The aggregate is unambiguous. Every queue item that votes, votes RE-OPEN or neutral; none votes to harden; the decisive (c)-scope computation proves that the E/V-artifact mechanism — the last candidate route by which the Wave 20 conditional FALSE could have been hardened — is vacuous on the analog object. The verdict therefore stands with Wave 21: P-δ_1 is re-opened, the primary-obstruction route to δ_1 = 3/2 is dead in the analog book (the d(F) pairing vanishes identically), and the degree-9 2-primary classes persist as genuine candidates. The bracket [4/3, 3/2] for δ_1 is unaffected; the secondary obstruction remains the unpinned frontier.

Two honest tensions must be recorded for the next wave. First, the analog count — twenty-three Z/2 summands at degree 9 — exceeds the W18 PD-mirror bound of at most one genuine class. The reconciliation is that the analog book is unbounded: it has no kappa-truncation, so the Γ-homology contributions accumulate without the wrapping that the n = 4 campaign's complex imposes. The decisive test of that reconciliation is to re-run the (c)-scope machinery on the 932-stratum n = 4 assembly itself, where the truncation is present and the [1, 3, 3, 1] pattern lives. Second, the V-book of Table 4 — the $(Z/4)^2$ at the middle degree — exists as a separate coinvariant layer that does not feed the orbit complex; whether the n = 4 object possesses an analogous orphan layer that was silently dropped from its book is a falsifiable question and should be checked before the re-open verdict is treated as final.

The next-wave directives, in priority order: (i) transfer the firing-census and provenance machinery to the n = 4 complex — label each of the 932 strata's coverings by character-book, compute the per-book firings, and read the degree-9 provenance table exactly as Table 6 does here; (ii) test the orphan-layer question — search the n = 4 book for coinvariant layers that do not feed the orbit complex, the analog of the ψ_ρ-book; (iii) pin the secondary obstruction $H^4(B_4; Z)$, which is now the only live route to a value of δ_1; and (iv) run the level-4 cross-check of the level-crossing certificate, since the present computation, like Waves 20 and 21, certifies levels 2 and 3 only. The qutrit verdict remains untouched ($H_2(B_3)$ = Z/3, δ_2 = 4/3), and the manuscripts are unaffected by this round.

Artifacts of record: scripts/w19_core.py (the verified model), scripts/w19_cscope.py (the decisive computation), scripts/w19_queue.py (the queue), with outputs w19_core_output.txt, w19_cscope_output.txt, w19_queue_output.txt and the JSON data files w19_cscope_data.json and w19_queue_data.json. All verification gates pass; every number quoted in this report is machine-computed by those scripts.